PROJECT DESIGN PHASE-I

Problem - Solution Fit Template

Date: 28 October 2026	**Team ID:** Individual Project
Project Name: Health360_HealthCare_Analysis	**Maximum Marks:** 2 Marks

1. CUSTOMER SEGMENT(S)

• Clinical Risk Assessment Officers (e.g., Dr. Michael Carter) evaluating disease vulnerability.
• Healthcare Data Analysts (e.g., Emily Rodriguez) stratifying multi-variable patient indicators.
• Public Health Researchers & Policy Planners (e.g., Dr. Priya Nair) planning preventive initiatives across 5,000 patient cohorts.

6. CUSTOMER CONSTRAINTS

• Time Constraints: Clinicians cannot manually review thousands of EHR spreadsheet rows.
• Data Complexity: Correlating clinical metrics (BMI, BP, Glucose) with lifestyle habits (Smoking, Alcohol, Exercise) requires unified visual tools.
• Resource Limitations: Public health budgets require targeted high-risk cohort identification.

5. AVAILABLE SOLUTIONS

• Static EMR/EHR Database Exports: Raw tables lacking real-time cross-filtering.
• Basic Excel Spreadsheets: Manual pivot tables unable to render complex multi-disease heatmaps.
• Fragmented Medical Logs: Isolated patient charts without population-level risk trend visibility.

2. JOBS-TO-BE-DONE / PROBLEMS

• Stratify 5,000 patient records across key chronic conditions (Diabetes 34.92%, Heart Disease 29.98%, Stroke 24.86%).
• Identify clinical indicators: Analyze Blood Pressure vs. Glucose scatter plots for early risk detection.
• Evaluate behavioral & demographic impact: Quantify disease prevalence across Age Groups (Middle Age to Senior) and BMI Categories (Obese 39.98% vs Normal 25.97%).

9. PROBLEM ROOT CAUSE

• Siloed Patient Data: Clinical metrics (BP, Glucose, Cholesterol) and lifestyle metrics (Smoking, Exercise Level 0-2) stored in disconnected databases.
• Lack of Interactive Visualization: Traditional reporting tools fail to provide instant dynamic cohort filtering.
• Late Detection: Reliance on manual inspection delays preventative interventions for high-risk patients.

7. BEHAVIOUR

• Analysts export batch CSVs and spend hours building manual summary charts.
• Clinicians review patients reactively during acute events rather than proactively analyzing risk cohorts.
• Public health teams rely on outdated aggregate census reports rather than interactive patient stories.

3. TRIGGERS & 4. EMOTIONS

3. TRIGGERS
• Sudden spikes in chronic disease readmissions (Diabetes, Heart Disease, Stroke).
• Annual public health audit requiring population risk stratification.

4. EMOTIONS: BEFORE / AFTER
• Before: Overwhelmed by dense clinical spreadsheets, anxious about missing critical early warnings.
• After: Confident, proactive, and empowered by interactive Tableau visual stories and instant risk filters.

10. YOUR SOLUTION

Health360 Tableau Healthcare Analytics Solution:
• Interactive 8-View Visual Story ('Story 1') integrating clinical metrics and lifestyle behaviors across 5,000 patients.
• Dynamic KPI Cards: Real-time tracking of Diabetes (34.92%), Heart Disease (29.98%), and Stroke (24.86%) rates.
• Multi-Variable Visual Analytics: BP vs. Glucose Scatter Plots, Age-Group Heatmaps, BMI Risk Histograms (Obese 39.98%), and Exercise-Level Pie Charts (Levels 0–2).
• Actionable Stratification: Instant hover tooltips and dynamic filters for early medical intervention.

8. CHANNELS OF BEHAVIOUR

8.1 ONLINE CHANNELS:
• Web-based Tableau Public / Server Interactive Dashboard Portal.
• EMR/EHR System Integration APIs pushing automated high-risk alerts.

8.2 OFFLINE CHANNELS:
• Executive Visual Story Presentations during clinical board reviews.
• Printable PDF Summary Reports for public health policy planning sessions.